

p-adic Numbers

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1 p -Adic Metric

Let K an ultrametric space, $x, y \in K$ st $|x| \neq |y|$. Then $|x + y| = \max\{|x|, |y|\}$.

First consequences. $|1| \cdot |1| = |1| \rightarrow |1| = 1$, since $1 \neq 0$. $|-1| \cdot |-1| = |1| = 1 \rightarrow |-1| = 1$. Then $|-x| = |-1| \cdot |x| = |x|$.

Result. WLOG $|x| < |y|$. Then $|x + y| \leq \max\{|x|, |y|\} = |y|$. We have $|y| \leq \max\{|x + y|, |x|\}$. Since $|y| > |x|$, we have $|x + y| \geq |x|$, so $|y| \leq |x + y|$, so $|x + y| = |y|$, as desired.

Define $v_p : \mathbb{Z} \rightarrow \mathbb{Z} \cup \infty$ by $v_p(0) = \infty$ and $v_p(x) = k$, where $x = p^k r$, $p \nmid r$. Note that $v_p(mn) = v_p(m) + v_p(n)$ for all $m, n \in \mathbb{Z}$. Define $v_p : \mathbb{Q} \rightarrow \mathbb{Z} \cup 0$ by $v_p(m/n) = v_p(m) - v_p(n)$ for $n \neq 0$.

I claim that v_p is well-defined. If $m = 0$, then $\infty - z = \infty$ for all $z \in \mathbb{Z}$, so $v_p(0) = \infty$. If $m \neq 0$, suppose $m/n = m'/n'$. Then $mn' = m'n$, so $v_p(m) + v_p(n') = v_p(m') + v_p(n)$, so $v_p(m) - v_p(n) = v_p(m') - v_p(n')$, so v_p is well-defined on $\mathbb{Q}$.

Suppose $r_1 = m_1/n_1, r_2 = m_2/n_2$, $m_1, m_2 \neq 0$. Then

$$\begin{aligned} v_p(r_1 r_2) &= v_p\left(\frac{m_1 m_2}{n_1 n_2}\right) = v_p(m_1 m_2) - v_p(n_1 n_2) \\ &= (v_p(m_1) + v_p(m_2)) - (v_p(n_1) + v_p(n_2)) = v_p(r_1) + v_p(r_2). \end{aligned}$$

If either m_1, m_2 is zero, then both sides are ∞ .

Note that $v_p(x + y) \geq \min\{v_p(x), v_p(y)\}$.

Define $|x| = \frac{1}{p^{v_p(x)}}$ for $x \in \mathbb{Q}$. Note that $|x| = 0$ iff $x = 0$. Note that

$$|xy| = \frac{1}{p^{v_p(xy)}} = \frac{1}{p^{v_p(x) + v_p(y)}} = \frac{1}{p^{v_p(x)}} \cdot \frac{1}{p^{v_p(y)}} = |x| \cdot |y|$$

WLOG $v_p(x) \leq v_p(y)$, so $|x| = \frac{1}{p^{v_p(x)}} \geq \frac{1}{p^{v_p(y)}} = |y|$ and

$$|x + y| = \frac{1}{p^{v_p(x+y)}} \leq \frac{1}{p^{v_p(x)}} = |x| = \max\{|x|, |y|\}.$$

Hence, $|\cdot|$ is a valid absolute value on $\mathbb{Q}$. Denote the induced ultrametric space by $\mathbb{Q}_p$.

Let (a_n) be a sequence of rationals. Say that (a_n) is p -Cauchy if (a_n) is Cauchy wrt $|\cdot|$.

Suppose $(a_n), (b_n)$ are Cauchy. Then I claim $(a_n + b_n)$ is Cauchy.

Let $\epsilon > 0$. Pick $N \in \mathbb{N}$ st $n, m \geq N \rightarrow |a_n - a_m|, |b_n - b_m| \leq \epsilon$. Then $|a_n + b_n - a_m - b_m| \leq \max\{|a_n - a_m|, |b_n - b_m|\} \leq \epsilon$.

Let $q \in \mathbb{Q}$. I claim that (qa_n) is Cauchy.

We have $|qa_n - qa_m| = |q| \cdot |a_n - a_m|$. If $q = 0$, we're done. Else, pick $N \in \mathbb{N}$ st $n, m \geq N \rightarrow |a_n - a_m| \leq \epsilon/|q|$ (note that this works since $|\cdot| : \mathbb{Q}_p \rightarrow \mathbb{R}_{\geq 0}$).

Corr. $(a_n - b_n)$ is Cauchy.

Suppose (a_n) is a $\mathbb{Q}$ -sequence. Say $a_n \rightarrow 0$ if for all $\epsilon > 0$, there exists $N \in \mathbb{N}$ st $n \geq N \rightarrow |a_n| \leq \epsilon$.

Sps $a_n \rightarrow 0$. I claim (a_n) is Cauchy. Indeed, pick $N \in \mathbb{N}$ st $n \geq N \rightarrow |a_n| \leq \epsilon$. Then for $m, n \geq N$, we have $|a_n - a_m| \leq \max\{|a_n|, |a_m|\} \leq \epsilon$.

Give $\mathbb{Q}^\omega$ the equivalence relation $\sim$ such that $(a_n) \sim (b_n)$ if $(a_n - b_n) \rightarrow 0$.

Note $(a_n) \sim (a_n)$ since $(0) \rightarrow 0$.

Sps. $(a_n) \sim (b_n)$. Since $|a_n - b_n| = |b_n - a_n|$, we have $(b_n) \sim (a_n)$.

Sps $(a_n) \sim (b_n) \sim (c_n)$. Let $\epsilon > 0$. Pick $N \in \mathbb{N}$ st $n \geq N \rightarrow |a_n - b_n| \leq \epsilon, |b_n - c_n| \leq \epsilon$. Then

$$|a_n - c_n| = |(a_n - b_n) + (b_n - c_n)| \leq \max\{|a_n - b_n|, |b_n - c_n|\} \leq \epsilon.$$

Hence, $(a_n) \sim (c_n)$, so $\sim$ is indeed an ER.

Let (a_n) be a $\mathbb{Q}$ -sequence. I claim that (a_n) is Cauchy iff for all $\epsilon > 0$, there exists $N \in \mathbb{N}$ st $n \geq N \rightarrow |a_n - a_{n+1}| \leq \epsilon$.

Proof: Let $\epsilon > 0$. Pick $N \in \mathbb{N}$ st $n \geq N \rightarrow |a_n - a_{n+1}| \leq \epsilon$. Then for $m \geq n \geq N$, we have $|a_n - a_{m+1}| \leq \max\{|a_n - a_m|, |a_m - a_{m+1}|\} \leq \epsilon$.

Sps $x = (x_n) \in \mathbb{Q}_p$. Define $|x| = \lim_{n \rightarrow \infty} |x_n|$. Let $\epsilon > 0$. Since (x_n) is Cauchy, pick $N \in \mathbb{N}$ st $n, m \geq N \rightarrow |x_n - x_m| \leq \epsilon$. Then for $n, m \geq N$, we have

$$||x_n| - |x_m|| \leq |x_n - x_m| \leq \epsilon,$$

so $(|x_n|)$ is Cauchy. Since $\mathbb{R}$ is complete, $|x|$ is well-defined.

Suppose $(x_n) \sim (y_n)$. Then I claim $|x| = |y|$. We have

$$||x_n| - |y_n|| \leq |x_n - y_n|,$$

so $|x| = |y|$. Hence, $|\cdot|$ is well-defined on $\mathbb{Q}_p$.

Let $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x| \leq 1\}$. I claim that $\mathbb{Z}_p$ is a ring.

Proof: We have $0 \in \mathbb{Z}_p$, so $\mathbb{Z}_p \neq \emptyset$. Sps $x, y \in \mathbb{Z}_p$. Then $|x - y| \leq \max\{|x|, |y|\} \leq 1$, so $x - y \in \mathbb{Z}_p$. Then $|xy| = |x| \cdot |y| \leq 1$, so $xy \in \mathbb{Z}_p$. Hence, $\mathbb{Z}_p$ is a ring.

For $z \in \mathbb{Z}$, note that $|z| \leq 1$ since $v_p(z) \geq 0$. Hence, we have $\mathbb{Z} \hookrightarrow \mathbb{Z}_p$. We also have $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$.

Claim: $\sum_{n=0}^{\infty} p^n = \frac{1}{1-p}$.

Proof: We have $s_N = \sum_{n=0}^N p^n = \frac{1-p^{N+1}}{1-p} = \frac{1}{1-p} - \frac{p^{N+1}}{1-p}$. Then

$$\left| s_N - \frac{1}{1-p} \right| \leq \frac{1}{p^{N+1}},$$

so $s_N \rightarrow \frac{1}{1-p}$ as $N \rightarrow \infty$.

Claim: Sps $\lim_{n \rightarrow \infty} a_n = a$. Then $\lim_{n \rightarrow \infty} |a_n| = |a|$.

Proof: We have $||a_n| - |a|| \leq |a_n - a|$.

Claim: Any series $\sum_{n=0}^{\infty} a_n p^n$ converges in $\mathbb{Q}_p$, where $a_n \in [p-1]$.

Proof: We have $|a_n p^n| = |a_n| \cdot |p^n| \leq \frac{1}{p^n}$. Since $1/p^n \rightarrow 0$ as $n \rightarrow \infty$, the series converges.

Claim: If $\sum_{n=n_0}^{\infty} a_n p^n = \sum_{m=m_0}^{\infty} b_m p^m$, then $n_0 = m_0$ and $a_n = b_n$ for all $n \geq n_0$.

Sps $\sum_{n=n_0}^{\infty} a_n p^n = \sum_{m=m_0}^{\infty} b_m p^m$. Sps FTSOC $n_0 \neq m_0$. If there exists $n_1 > n_0$

such that $a_{n_1} \neq 0$, then

$$\left| \sum_{n=n_1}^{\infty} a_n p^n \right| = \lim_{N \rightarrow \infty} \left| \sum_{n=n_1}^N a_n p^n \right| \leq \lim_{N \rightarrow \infty} |a_{n_1} p^{n_1}| = \frac{1}{p^{n_1}}.$$

We have $|a_{n_0} p^{n_0}| = \frac{1}{p^{n_0}} > \frac{1}{p^{n_1}}$, so

$$\left| \sum_{n=n_0}^{\infty} a_n p^n \right| = \frac{1}{p^{n_0}}.$$

If there does not exist such n_1 , the above equality still holds.

We have a similar result on $\sum_{m=m_0}^{\infty} b_m p^m$, so $1/p^{n_0} = 1/p^{m_0} \rightarrow n_0 = m_0$.

Sps FTSOC $a_{n_0} > b_{n_0}$. Let $d = p - a_{n_0}$. Note that $b_{n_0} + d \in [[p - 1]$. If $a_n = p - 1$ for all $n > n_0$, we have after adding $d p^{n_0}$ to the LHS,

$$(a_{n_0} + d)p^{n_0} + \sum_{n=n_0+1}^{\infty} (p-1)p^n = p^{n_0+1} + \sum_{n=n_0+1}^{\infty} (p-1)p^n = 0,$$

while the RHS is

$$(b_{n_0} + d)p^{n_0} + \sum_{n=n_0+1}^{\infty} b_n p^n.$$

Taking absolute values gives $0 = 1/p^{n_0}$, absurd.

Sps there exists $n_1 \geq n_0$ such that $a_{n_1+1} \neq p - 1$. Then we have for the LHS

$$\begin{aligned} & (a_{n_0} + d)p^{n_0} + \sum_{n=n_0+1}^{n_1} (p-1)p^n + \sum_{n=n_1+1}^{\infty} a_n p^n \\ &= (a_{n_1+1} + 1)p^{n_1+1} + \sum_{n=n_1+2}^{\infty} a_n p^n \end{aligned}$$

and for the RHS:

$$(b_{n_0} + d)p^{n_0} + \sum_{n=n_0+1}^{\infty} b_n p^n.$$

Comparing absolute values, we have $1/p^{n_1+1} = 1/p^{n_0}$, but

$$\frac{1}{p^{n_1+1}} \leq \frac{1}{p^{n_0+1}} < \frac{1}{p^{n_0}},$$

absurd.

Hence, we have $a_{n_0} = b_{n_0}$. Subtracting $a_{n_0}p^{n_0}$ from both sides, by induction we have $a_n = b_n$ for all $n \geq n_0$.

Claim: Let $\alpha \in \mathbb{Z}_p$ with $\alpha = \sum_{n=0}^{\infty} a_n p^n$. Then $\alpha \equiv \sum_{n=0}^{k-1} a_n p^n \pmod{p^k}$ for all $k \in \mathbb{N}$. Note that since $\alpha \in \mathbb{Z}_p$, we have $n_0 \geq 0$, so the LHS is reasonable.

Proof: We have $\alpha - \sum_{n=0}^{k-1} a_n p^n = \sum_{n=k}^{\infty} a_n p^n = p^k \sum_{n=0}^{\infty} a_{n+k} p^n \in p^k \mathbb{Z}_p$, so the result holds.

Sps $3x_n \equiv 4 \pmod{5^n}$. Sps $3x_{n+1} \equiv 4 \pmod{5^{n+1}}$. Write $x_{n+1} = 5^n q + x_n$. Then $3(5^n q + x_n) \equiv 4 \pmod{5^{n+1}}$, so $3q \equiv -\frac{3x_n-4}{5^n} \pmod{5}$, so $q \equiv 3 \cdot \frac{3x_n-4}{5^n} \pmod{5}$, so $x_{n+1} \equiv 5^n \cdot 3 \cdot \frac{3x_n-4}{5^n} + x_n = 10x_n - 12$.

$x \in \mathbb{Q}_p$ is eventually periodic iff $x \in \mathbb{Q}$.

Consider $a, b \in \mathbb{Z}$ with $a, b \perp p$. Consider the equation $bx_1 \equiv a \pmod{p}$, so $x_1 \equiv b^{-1}a \pmod{p}$. Consider $bx_{n+1} \equiv a \pmod{p^{n+1}}$. Write $x_{n+1} = p^n q_{n+1} + x_n$. We have $b(p^n q_{n+1} + x_n) \equiv a \pmod{p^{n+1}} \rightarrow bp^n q_{n+1} \equiv -(bx_n - a) \pmod{p^{n+1}} \rightarrow q_{n+1} \equiv -b^{-1} \frac{bx_n - a}{p^n} \pmod{p}$. Note that b^{-1} is taken modulo p (and is independent of n).

Hence, we have $x_{n+1} \equiv -b^{-1}(bx_n - a) + x_n \equiv (1 - b^{-1}b)x_n + b^{-1}a \pmod{p^{n+1}}$. Note that modulo p^n , we have $x_{n+1} \equiv x_n - b^{-1}a + b^{-1}a \equiv x_n \pmod{p^n}$.

Hence, we have a well-defined sequence $r_n \in [[p-1]]$ of residues such that $b \sum_{n=0}^{k-1} r_k p^k \equiv a \pmod{p^k}$. In other words,

$$\left| b \sum_{n=0}^{k-1} r_k p^k - a \right| \leq \frac{1}{p^k} \rightarrow \left| \sum_{k=0}^{k-1} r_k p^k - \frac{a}{b} \right| \leq \frac{1}{p^k},$$

since $|b| = 1$ (since $b \perp p$).

It follows that $\frac{a}{b} = \sum_{n=0}^{\infty} r_n p^n$. We can scale a, b so that every nonzero rational has a proper expansion....? Some PHP argument to finish?

2 Topology on $\mathbb{Q}_p$

Let $x \in \mathbb{Q}_p$ and $\epsilon > 0$. Let $n \in \mathbb{Z}$ be minimal such that $p^{-n} < \epsilon$. I claim that $B(x, \epsilon) = \overline{B}(x, p^{-n})$.

Proof: We have $\overline{B}(x, p^{-n}) \subset B(x, \epsilon)$. Sps $y \in B(x, \epsilon)$. Then $|y - x| < \epsilon$.

Note that $|y - x| = p^{-m}$ for some $m \in \mathbb{Z}$, so $|y - x| \leq p^{-n}$, so $y \in \overline{B}(x, p^{-n})$. Hence, balls in $\mathbb{Q}_p$ are both open and closed.

Claim: $\mathbb{Z}_p$ is compact.

Proof: It suffices to show $\mathbb{Z}_p$ is sequentially compact.

Let (x_n) be bounded in $\mathbb{Z}_p$, so (x_n) is any sequence (since $|x_n| \leq 1$). Sps for all $\epsilon > 0$, there exists $n \in N$ st $|x_n| \leq \epsilon$. Then pick a subsequence converging to $0 \in \mathbb{Z}_p$.

Else, sps there exists $\epsilon > 0$ such that for all $n \in N$, we have $|x_n| \geq \epsilon$. Pick $m \in \mathbb{Z}_{\geq 0}$ such that $|x_n| \geq p^{-m}$, so $v_p(x$